

CYBERSECURITY PORTFOLIO

HANDS-ON LAB · LINUX SECURITY

Linux Server Hardening

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Portfolio Edition

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Original technical content and evidence preserved.
Other participant names removed for public portfolio use.

Week 2 Day 2 Lab



Part 1: System Updates and Patch Management

Check what updates are available:

Apply all available updates:

```
student@LinuxTargetVM3:~$ sudo apt update && apt list --upgradable
[sudo] password for student:
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease
Get:5 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1190 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [282 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1683 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [335 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1424 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [323 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [12.6 kB]
Fetched 5376 kB in 1s (3667 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Listing... Done
```

Write down what got updated, including anything related to the kernel.

```
student@LinuxTargetVM3:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

Check if a reboot is needed to finish applying patches:

If this file exists and prints a message, note it in your findings. Do not reboot the machine yourself. Flag it to your instructor first.

```
student@LinuxTargetVM3:~$ cat /var/run/reboot-required
*** System restart required ***
```

Install unattended-upgrades so future security patches apply automatically:

Confirm it's enabled:

```
student@LinuxTargetVM3:~$ systemctl status unattended-upgrades
● unattended-upgrades.service - Unattended Upgrades Shutdown
   Loaded: loaded (/usr/lib/systemd/system/unattended-upgrades.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-08-05 17:05:58 UTC; 6 days ago
     Docs: man:unattended-upgrade(8)
  Main PID: 1087 (unattended-upgr)
    Tasks: 2 (limit: 9434)
   Memory: 9.4M (peak: 9.5M)
      CPU: 131ms
   CGroup: /system.slice/unattended-upgrades.service
           └─1087 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal

Aug 05 17:05:58 LinuxTargetVM3 systemd[1]: Started unattended-upgrades.service - Unattended Upgrades Shutdown.
student@LinuxTargetVM3:~$ |
```

Part 2: Disabling Unused Services and Ports

Running services:

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ systemctl list-units --type=service --state=running
```

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
auditd.service	loaded	active	running	Security Auditing Service
chrony.service	loaded	active	running	chrony, an NTP client/server
cron.service	loaded	active	running	Regular background program processing daemon
dbus.service	loaded	active	running	D-Bus System Message Bus
fail2ban.service	loaded	active	running	Fail2Ban Service
fwupd.service	loaded	active	running	Firmware update daemon
getty@tty1.service	loaded	active	running	Getty on tty1
hv-kvp-daemon.service	loaded	active	running	Hyper-V KVP Protocol Daemon
inetutils-inetd.service	loaded	active	running	GNU Network Utilities internet superserver
ModemManager.service	loaded	active	running	Modem Manager
multipathd.service	loaded	active	running	Device-Mapper Multipath Device Controller
networkd-dispatcher.service	loaded	active	running	Dispatcher daemon for systemd-networkd
packagekit.service	loaded	active	running	PackageKit Daemon
polkit.service	loaded	active	running	Authorization Manager
postfix@-.service	loaded	active	running	Postfix Mail Transport Agent (instance -)
rsyslog.service	loaded	active	running	System Logging Service
serial-getty@ttyS0.service	loaded	active	running	Serial Getty on ttyS0
ssh.service	loaded	active	running	OpenBSD Secure Shell server
systemd-journald.service	loaded	active	running	Journal Service
systemd-logind.service	loaded	active	running	User Login Management
systemd-networkd.service	loaded	active	running	Network Configuration
systemd-resolved.service	loaded	active	running	Network Name Resolution
systemd-udev.service	loaded	active	running	Rule-based Manager for Device Events and Files
udisks2.service	loaded	active	running	Disk Manager
unattended-upgrades.service	loaded	active	running	Unattended Upgrades Shutdown
user@1000.service	loaded	active	running	User Manager for UID 1000
walinuxagent.service	loaded	active	running	Azure Linux Agent

```
lines 1-29...skipping...
```

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
auditd.service	loaded	active	running	Security Auditing Service
chrony.service	loaded	active	running	chrony, an NTP client/server
cron.service	loaded	active	running	Regular background program processing daemon
dbus.service	loaded	active	running	D-Bus System Message Bus
fail2ban.service	loaded	active	running	Fail2Ban Service
fwupd.service	loaded	active	running	Firmware update daemon
getty@tty1.service	loaded	active	running	Getty on tty1
hv-kvp-daemon.service	loaded	active	running	Hyper-V KVP Protocol Daemon
inetutils-inetd.service	loaded	active	running	GNU Network Utilities internet superserver
ModemManager.service	loaded	active	running	Modem Manager
multipathd.service	loaded	active	running	Device-Mapper Multipath Device Controller
networkd-dispatcher.service	loaded	active	running	Dispatcher daemon for systemd-networkd
packagekit.service	loaded	active	running	PackageKit Daemon
polkit.service	loaded	active	running	Authorization Manager

postfix@-.service	loaded	active	running	Postfix Mail Transport Agent (instance -)
rsyslog.service	loaded	active	running	System Logging Service
serial-getty@ttyS0.service	loaded	active	running	Serial Getty on ttyS0
ssh.service	loaded	active	running	OpenBSD Secure Shell server
systemd-journald.service	loaded	active	running	Journal Service
systemd-logind.service	loaded	active	running	User Login Management
systemd-networkd.service	loaded	active	running	Network Configuration
systemd-resolved.service	loaded	active	running	Network Name Resolution
systemd-udevd.service	loaded	active	running	Rule-based Manager for Device Events and Files
udisks2.service	loaded	active	running	Disk Manager
unattended-upgrades.service	loaded	active	running	Unattended Upgrades Shutdown
user@1000.service	loaded	active	running	User Manager for UID 1000
walinuxagent.service	loaded	active	running	Azure Linux Agent

Legend: LOAD → Reflects whether the unit definition was properly loaded.
ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
SUB → The low-level unit activation state, values depend on unit type.

27 loaded units listed.

2 2 2 2 2 2 2

None

```
systemctl list-units --type=service --state=running
```

Listening ports:

```
student@linuxtargetVM3:~$ pwd
/home/student
student@linuxtargetVM3:~$ sudo ss -tulnp
Netid      State     Recv-Q     Send-Q           Local Address:Port               Peer Address:Port        Process
udp        UNCONN     0          0           127.0.0.54:53                   0.0.0.0:*                 users(("systemd-resolve",pid=128737,fid=16))
udp        UNCONN     0          0           127.0.0.53:101:53              0.0.0.0:*                 users(("systemd-resolve",pid=128737,fid=16))
udp        UNCONN     0          0           172.16.0.4#eth0:68             0.0.0.0:*                 users(("systemd-network",pid=128580,fid=22))
udp        UNCONN     0          0           127.0.0.1:1:323               0.0.0.0:*                 users(("chronyd",pid=1086,fid=6))
udp        UNCONN     0          0           [::]:1:323                     [::]:*                   users(("chronyd",pid=1086,fid=7))
tcp        LISTEN     0          4096       127.0.0.54:53                   0.0.0.0:*                 users(("systemd-resolve",pid=128737,fid=17))
tcp        LISTEN     0          100      0.0.0.0:0:25                   0.0.0.0:*                 users(("master",pid=132927,fid=13))
tcp        LISTEN     0          10      0.0.0.0:0:23                   0.0.0.0:*                 users(("inetutils-inetd",pid=147481,fid=4))
tcp        LISTEN     0          4096       127.0.0.22:22                  0.0.0.0:*                 users(("sshd",pid=112304,fid=4))
tcp        LISTEN     0          4096       127.0.0.53:101:53              0.0.0.0:*                 users(("systemd-resolve",pid=128737,fid=15))
tcp        LISTEN     0          100      [::]:1:25                     [::]:*                   users(("master",pid=132927,fid=14))
tcp        LISTEN     0          4096       [::]:22                       [::]:*                   users(("sshd",pid=112304,fid=4),("systemd",pid=1,fid=117))
```

None

```
sudo ss -tulpn
```

Service/Port	Keep or Disable	Reasoning
systemd-resolved — DNS, port 53 on 127.0.0.53/54	Keep	Provides local DNS resolution. The addresses are loopback-only (127.0.0.x), so this is not exposed externally.
systemd-networkd — DHCP client, UDP 68	Keep	Handles network configuration/DHCP for the VM. The server needs this for network connectivity.
chronyd — NTP, UDP 323 on 127.0.0.1	Keep	Maintains accurate system time, which is important for logs, auditing, authentication, and certificates.
Postfix — SMTP, TCP 25	Disable	Provides SMTP mail service. A banking web/file server does not normally need to act as a mail server, so this is unnecessary exposure.
inetutils-inetd / Telnet — TCP 23	Keep	
sshd — SSH, TCP 22	Keep	Provides secure remote administration. This is needed to manage the server remotely.
Systemd — TCP 22	Keep	This appears alongside SSH because systemd is involved in managing/socket activation for the SSH service. It is not a separate network service you would disable here.
ModemManager.service — No Port Listed	Disable	ModemManager was found running in the systemctl service list, but it does not appear in ss -tuln because it is not currently listening on a TCP or UDP port. An Azure banking web/file server does not need cellular modem management, so the service is unnecessary and can be disabled to reduce the system's attack surface.

Disable:

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ sudo systemctl disable --now ModemManager.service
[sudo] password for student:
Removed "/etc/systemd/system/multi-user.target.wants/ModemManager.service".
Removed "/etc/systemd/system/dbus-org.freedesktop.ModemManager1.service".
student@LinuxTargetVM3:~$ sudo systemctl disable --now postfix.service
Synchronizing state of postfix.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable postfix
Removed "/etc/systemd/system/multi-user.target.wants/postfix.service".
```

None

```
sudo systemctl disable --now <service-name>
```

Confirmation:

None

```
sudo ss -tulpn
```

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ sudo ss -tulpn
```

Netid	State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port	Process
udp	UNCONN	0	0	127.0.0.54:53	0.0.0.0:*	users:(("systemd-resolve",pid=128737,fd=16))
udp	UNCONN	0	0	127.0.0.53%lo:53	0.0.0.0:*	users:(("systemd-resolve",pid=128737,fd=14))
udp	UNCONN	0	0	172.16.0.4%eth0:68	0.0.0.0:*	users:(("systemd-network",pid=128580,fd=22))
udp	UNCONN	0	0	127.0.0.1:323	0.0.0.0:*	users:(("chronyd",pid=1086,fd=6))
udp	UNCONN	0	0	:::1:323	:::1:*	users:(("chronyd",pid=1086,fd=7))
tcp	LISTEN	0	4096	127.0.0.54:53	0.0.0.0:*	users:(("systemd-resolve",pid=128737,fd=17))
tcp	LISTEN	0	10	0.0.0.0:23	0.0.0.0:*	users:(("inetutils-inetd",pid=147481,fd=4))
tcp	LISTEN	0	4096	0.0.0.0:22	0.0.0.0:*	users:(("sshd",pid=112384,fd=3),("systemd",pid=1,fd=113))
tcp	LISTEN	0	4096	127.0.0.53%lo:53	0.0.0.0:*	users:(("systemd-resolve",pid=128737,fd=15))
tcp	LISTEN	0	4096	:::22	:::1:*	users:(("sshd",pid=112384,fd=4),("systemd",pid=1,fd=115))

```
student@LinuxTargetVM3:~$ systemctl status ModemManager.service postfix.service
o ModemManager.service - Modem Manager
   Loaded: loaded (/usr/lib/systemd/system/ModemManager.service; disabled; preset: enabled)
   Active: inactive (dead)

Aug 11 06:04:33 LinuxTargetVM3 systemd[1]: ModemManager.service: Deactivated successfully.
Aug 11 06:04:33 LinuxTargetVM3 systemd[1]: Stopped ModemManager.service - Modem Manager.
Aug 11 06:04:33 LinuxTargetVM3 systemd[1]: Starting ModemManager.service - Modem Manager...
Aug 11 06:04:33 LinuxTargetVM3 ModemManager[132545]: <msg> ModemManager (version 1.23.4) starting in system bus...
Aug 11 06:04:33 LinuxTargetVM3 systemd[1]: Started ModemManager.service - Modem Manager.
Aug 11 18:52:02 LinuxTargetVM3 ModemManager[132545]: <msg> caught signal, shutting down...
Aug 11 18:52:02 LinuxTargetVM3 systemd[1]: Stopping ModemManager.service - Modem Manager...
Aug 11 18:52:02 LinuxTargetVM3 ModemManager[132545]: <msg> ModemManager is shut down
Aug 11 18:52:02 LinuxTargetVM3 systemd[1]: ModemManager.service: Deactivated successfully.
Aug 11 18:52:02 LinuxTargetVM3 systemd[1]: Stopped ModemManager.service - Modem Manager.

o postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; disabled; preset: enabled)
   Active: inactive (dead)
   Docs: man:postfix(1)

Aug 10 18:21:31 LinuxTargetVM3 systemd[1]: Starting postfix.service - Postfix Mail Transport Agent...
Aug 10 18:21:31 LinuxTargetVM3 systemd[1]: Finished postfix.service - Postfix Mail Transport Agent.
Aug 11 18:52:12 LinuxTargetVM3 systemd[1]: postfix.service: Deactivated successfully.
Aug 11 18:52:12 LinuxTargetVM3 systemd[1]: Stopped postfix.service - Postfix Mail Transport Agent.
```


Part 3: Kernel Hardening with sysctl

1. Review current network-related kernel settings:

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ sysctl -a | grep net.ipv4
sysctl: permission denied on key 'kernel.apparmor_display_secid_mode'
sysctl: permission denied on key 'kernel.apparmor_restrict_unprivileged_io_uring'
sysctl: permission denied on key 'kernel.apparmor_restrict_unprivileged_userns_complain'
sysctl: permission denied on key 'kernel.apparmor_restrict_unprivileged_userns_force'
sysctl: permission denied on key 'kernel.cad_pid'
sysctl: permission denied on key 'kernel.unprivileged_userns_apparmor_policy'
sysctl: permission denied on key 'kernel.usermodehelper.bset'
sysctl: permission denied on key 'kernel.usermodehelper.inheritable'
sysctl: permission denied on key 'net.core.bpf_jit_harden'
sysctl: permission denied on key 'net.core.bpf_jit_kallsyms'
sysctl: permission denied on key 'net.core.bpf_jit_limit'
net.ipv4.cipso_cache_bucket_size = 10
net.ipv4.cipso_cache_enable = 1
net.ipv4.cipso_rbm_optfmt = 0
net.ipv4.cipso_rbm_strictvalid = 1
net.ipv4.conf.all.accept_local = 0
net.ipv4.conf.all.accept_redirects = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.all.arp_accept = 0
net.ipv4.conf.all.arp_announce = 0
net.ipv4.conf.all.arp_evict_nocarrier = 1
net.ipv4.conf.all.arp_filter = 0
net.ipv4.conf.all.arp_ignore = 0
net.ipv4.conf.all.arp_notify = 0
net.ipv4.conf.all.bc_forwarding = 0
net.ipv4.conf.all.bootp_relay = 0
net.ipv4.conf.all.disable_policy = 0
net.ipv4.conf.all.disable_xfrm = 0
net.ipv4.conf.all.drop_gratuitous_arp = 0
net.ipv4.conf.all.drop_unicast_in_l2_multicast = 0
net.ipv4.conf.all.force_igmp_version = 0
net.ipv4.conf.all.forwarding = 0
net.ipv4.conf.all.igmpv2_unsolicited_report_interval = 10000
net.ipv4.conf.all.igmpv3_unsolicited_report_interval = 1000
net.ipv4.conf.all.ignore_routes_with_linkdown = 0
net.ipv4.conf.all.log_martians = 0
net.ipv4.conf.all.mc_forwarding = 0
net.ipv4.conf.all.medium_id = 0
net.ipv4.conf.all.promote_secondaries = 0
net.ipv4.conf.all.proxy_arp = 0
net.ipv4.conf.all.proxy_arp_pvlan = 0
net.ipv4.conf.all.route_localnet = 0
net.ipv4.conf.all.rp_filter = 2
```

None

```
sysctl -a | grep net.ipv4
```

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ sudo nano /etc/sysctl.d/99-hardening.conf
student@LinuxTargetVM3:~$ |
```

None

```
sudo nano /etc/sysctl.d/99-hardening.conf
```

Lines before:

```
net.ipv4.tcp_syncookies = 1
```

```
net.ipv4.ip_forward = 0
```

```
net.ipv4.icmp_echo_ignore_broadcasts = 1
```

```
net.ipv4.conf.all.accept_source_route = 0
```

```
net.ipv4.conf.default.accept_source_route = 1
```

Add these lines:

```
GNU nano 7.2 /etc/sysctl.d/99-hardening.conf
net.ipv4.ip_forward = 0
net.ipv4.icmp_echo_ignore_broadcasts = 1
net.ipv4.tcp_syncookies = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.default.accept_source_route = 0
[ Wrote 5 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^N Replace    ^U Paste      ^J Justify
^C Location  ^M-U Undo     ^M-A Set Mark
^_ Go To Line ^M-E Redo     ^M-G Copy
```

None

```
net.ipv4.ip_forward = 0
net.ipv4.icmp_echo_ignore_broadcasts = 1
net.ipv4.tcp_syncookies = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.default.accept_source_route = 0
```

Apply the changes:

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ sudo sysctl --system
* Applying /usr/lib/sysctl.d/10-apparmor.conf ...
* Applying /etc/sysctl.d/10-bufferbloat.conf ...
* Applying /etc/sysctl.d/10-console-messages.conf ...
* Applying /etc/sysctl.d/10-ipv6-privacy.conf ...
* Applying /etc/sysctl.d/10-kernel-hardening.conf ...
* Applying /etc/sysctl.d/10-magic-sysrq.conf ...
* Applying /etc/sysctl.d/10-map-count.conf ...
* Applying /etc/sysctl.d/10-network-security.conf ...
* Applying /etc/sysctl.d/10-ptrace.conf ...
* Applying /etc/sysctl.d/10-zero-page.conf ...
* Applying /usr/lib/sysctl.d/50-pid-max.conf ...
* Applying /etc/sysctl.d/99-cloudimg-ipv6.conf ...
* Applying /etc/sysctl.d/99-cloudimg-udp.conf ...
* Applying /etc/sysctl.d/99-hardening.conf ...
* Applying /usr/lib/sysctl.d/99-protect-links.conf ...
* Applying /etc/sysctl.d/99-sysctl.conf ...
* Applying /etc/sysctl.conf ...
kernel.apparmor_restrict_unprivileged_userns = 1
net.core.default_qdisc = fq_codel
kernel.printk = 4 4 1 7
net.ipv6.conf.all.use_tempaddr = 2
net.ipv6.conf.default.use_tempaddr = 2
kernel.kptr_restrict = 1
kernel.sysrq = 176
vm.max_map_count = 1048576
net.ipv4.conf.default.rp_filter = 2
```

None

```
sudo sysctl --system
```

```
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.default.accept_source_route = 0
fs.protected_fifos = 1
fs.protected_hardlinks = 1
```

None

```
sysctl net.ipv4.ip_forward
sysctl net.ipv4.tcp_syncookies
```

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ cat /etc/sysctl.d/99-hardening.conf
net.ipv4.ip_forward = 0
net.ipv4.icmp_echo_ignore_broadcasts = 1
net.ipv4.tcp_syncookies = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.default.accept_source_route = 0
```

This is what the /etc/sysctl.d/99-hardening.conf file includes.

5. Hardening Summary

The Linux host was hardened by applying all available system and security updates to reduce exposure to any known vulnerabilities. Unattended upgrades were also configured so future security patches could be applied automatically.

We reviewed active services and listening ports and determined that Postfix and ModemManager were deemed unnecessary for the banking web/file server role, so both of those services were disabled to reduce the attack surface of the system. Telnet was identified as insecure but was intentionally left enabled since it was required for a later portion of the lab.

6. Reflection Questions

1. Which change today felt riskiest to make on a production-bound host, and how did you validate it didn't break anything?

The riskiest change made today was most likely the one involving telnet. As a weak application is can cause issues just by enabling it. Even though it was not totally enforced and verified it still caused issues that would need to be validated that it did not break the machine.

2. Why might disabling a service be a better fix than just blocking its port with a firewall rule? Are there cases where you'd want both?

Disabling a service reduces the attack surface as well as exposure. A firewall can fail or get bypassed so it adds another layer of security also known as defense in depth. In most real world settings both are the common setup for services you would need because of the added layers of defense. An example would be if a public web server has to be reachable due to legacy software. The firewall can be configured to allow only http and https and deny everything else.

3. How does mandatory access control (AppArmor) protect a host differently than the least-privilege user/file permissions work from Week 1

Least-privilege access permissions are used to control what users and groups are allowed to access on the system. AppArmor is able to go a step further by restricting what specific applications are allowed to do. Even if a user has permission to access something, AppArmor would still block them from accessing it. All this does is add another layer of security that helps limit any damage if a specific application is compromised. This goes into accordance of defense in depth by limiting both user privileges and application behavior.